Change of Base Formula & Evaluation - Nathaniel Juan P

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Student

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Topic: Change of Base Formula & Evaluation

Status: completed

Lesson Plan

Learning Objectives

- Apply the change of base formula to transform logarithmic expressions into common or natural logarithms.
- Calculate the numerical values of logarithms with non-standard bases using a scientific calculator.
- Compare the magnitude of different logarithmic expressions by converting them to a uniform base.
- Solve complex logarithmic equations that require base conversion as an intermediate step.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

> Hi Nathaniel! Great to see you today. Technology and engineering often require us to handle data across different scales, and logarithms are the tool for that. Before we dive into new territory, let's sharpen our tools. I've put 4 quick challenges on the whiteboard. Two are about the log properties we mastered last time, and two involve basic evaluation. Give these a shot in the chatbox!

Activities:

- *Solve: log,(8) + log,(4) using the product property.*
- *Expand: logf(x/y) using the quotient property.*
- *Evaluate log…(25) without a calculator.*
- *Estimate if log,(10) is closer to 3 or 4.*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Think about sophisticated sensors in engineering—like an earthquake seismograph or a pH meter. These devices often use different 'bases' for their calculations. However, most calculators only have buttons for 'log' (base 10) and 'ln' (base e). What happens if your sensor gives you data in base 7? Today, we learn the 'universal translator' of math: the Change of Base Formula. It allows us to calculate any logarithm using only the standard buttons on our calculator.

Activities:

- *Discussion on the 'log' and 'ln' buttons on a standard calculator.*
- *Visualizing a 'translator' metaphor for base changes.*

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Refresh your memory: a logarithm asks 'what exponent do I need?' For log_b(a) = c, it means b to the power of c equals a. If we can't find a clean power—like logƒ(10)—we a
Does anyone remember the structure of the Change of Base formula from your pre-reading?

Activities:

- *Writing the general form: log_b(a) = log_x(a) / log_x(b).*
- *Identifying 'b' as the old base and 'a' as the argument.*

4. Problem Introduction (10 minutes)

Teacher Script:

Let's tackle a 'hard' scenario. Imagine you are analyzing an exponential growth rate in a system using base 1.5, and you need to solve log_1.5(20). Your calculator doesn't have a base 1.5 button. Let's use the formula: log_1.5(20) = log(20) / log(1.5). Let's see how changing to base 10 (common log) or base 'e' (natural log) gives us the exact same answer.

Activities:

- *Teacher models the step-by-step conversion.*
- *Demonstration of calculating log(20)/log(1.5) vs ln(20)/ln(1.5).*

5. Guided Practice (15 minutes)

Teacher Script:

Now, Nathaniel, let's try three together. I'll set the stage, and you drive the calculator. First, evaluate log‡(50). Next, let's try a tricky one: 1 / log…(10). Finally, let's compa
logƒ(8) or log…(24)? We will convert both to base 10 to decide.

Activities:

- *Compute log‡(50) to 4 decimal places.*
- *Simplify 1 / (log 10 / log 5) and discuss the reciprocal property.*
- *Comparison task using base 10 conversion.*

6. Independent Practice (15 minutes)

Teacher Script:

Time for the 'Solo Flight.' I've listed 4 problems ranging from standard evaluations to comparing complex log expressions. Work through these. If you hit a wall, tell me which 'base' you are trying to switch to.

Activities:

- *Task 1: Evaluate log_0.5(12).*
- *Task 2: Solve for x: 3^x = 17 using logarithms and change of base.*
- *Task 3: Evaluate (log, 5)(log… 8) using conversion.*
- *Task 4: Order from least to greatest: log,(7), logƒ(20), log„(30).*

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Brilliant work today, Nathaniel. You've mastered how to bypass the limitations of your calculator! Remember: the old base always goes to the basement (the denominator). Before you go, please log in to https://learn.algonova.id/login to view your customized homework set. Any questions on the 'basement' rule?

Activities:

- *Summary of the 'Basement Rule' (Base goes to denominator).*
- *Direct link sharing for the portal.*

Homework

Medium Questions:

1. Evaluate log†(200) to three decimal places.
2. Evaluate log_1.2(5) using the natural logarithm (ln).
3. Use the change of base formula to rewrite logƒ(x) in terms of
4. Calculate the value of log…(0.2) using a calculator and verify

Hard Questions:

1. Find the exact value of (logƒ 7) × (log‡ 9) without a calculato
formula.
2. Solve for x to four decimal places: 5^(x-1) = 12.
3. Compare log, 10 and logƒ 30. Which is larger? Show yourWork
10.
4. Simplify the expression [log_a(b)] * [log_b(c)] * [log_c(d)] into a single logarithm.
5. If logˆ(3) = k, express logƒ(2) in terms of k.
6. Evaluate log_(" 2) (8) using the change of base formula.

Answer Key:

`log†(200) = log(200)/log(6) "H 2.956`
`log_1.2(5) = ln(5)/ln(1.2) "H 8.827`
`logƒ(x) = log‰(x)/log‰(3) = log‰(x)/0.5 = 2log‰(x)`
`log…(0.2) = log(0.2)/log(5) = -1`
`(log 7 / log 3) * (log 9 / log 7) = log 9 / log 3 = 2`
`(x-1)log 5 = log 12 -> x = (log 12 / log 5) + 1 "H 2.5440`
`log, 10 "H 3.32, logƒ 30 "H 3.10. Therefore log, 10 is larger.`
`(log b/log a)*(log c/log b)*(log d/log c) = log d / log a = log_a(d)`
`k = log 3 / log 8 = log 3 / 3log 2. Therefore log 3 / log 2 = 3k. l
`log(8)/log(" 2) = 3log 2 / 0.5log 2 = 6`
REVIEW: Properties of Logarithms - `log,(16*4) = log, 16 + log, 4
REVIEW: Properties of Logarithms - `log 1000 - log 10 = log(1000/10) = log 100 = 2`
REVIEW: Properties of Logarithms - `log $x^3$ = 3 log x`
REVIEW: Properties of Logarithms - `log,(1/8) = -3`

Parent Reminder

Today's Math Progress: Nathaniel

Nathaniel mastered the 'Change of Base' formula today! This is a high-level skill that allows him to solve logarithmic problems that standard calculators can't handle directly. He worked through complex comparisons and practiced using natural logarithms. Please encourage him to complete the homework on the portal to reinforce these multi-step calculations.